

IBM SKILLSBUILD PROJECT

HireMate AI

Intelligent Job Recommendation Agent

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Academic Year: 2026–2027



THE CHALLENGE

Problem Statement & Objectives

Challenges Faced by Students & Job Seekers

1

Lack of Personalized Guidance

Students receive generic career advice that does not account for their unique skill sets, interests, or academic background.

2

Skill Mismatches & Gaps

Job seekers struggle to identify which skills are in demand and how their current abilities align with employer expectations.

3

Resume & Interview Unpreparedness

Many candidates fail to optimize their resumes for ATS systems or adequately prepare for technical and HR interviews.

Project Objectives

- Recommend suitable job roles
- Analyze skills & identify skill gaps
- Suggest certifications & learning resources
- Generate personalized learning roadmaps
- Provide resume & interview guidance
- Improve employability via AI-powered assistance

About HireMate AI & Methodology

HireMate AI is an intelligent AI agent built using **IBM watsonx Orchestrate** that analyzes user profiles and provides fully personalized, end-to-end career guidance — from job recommendations to interview preparation.



The agent follows a structured pipeline — starting from a user's profile and systematically generating actionable career intelligence at every stage of the job-seeking journey.

Features, Use Cases & Benefits

Key Features



Personalized Job Recommendations



Skill Gap Analysis



Learning Roadmap Generation



Resume Suggestions



Interview Preparation



Certification Recommendations

Use Cases



Campus Placements



Internship Search



Career Switching



Fresh Graduate Guidance



Skill Development

Why AI Agent vs. Traditional Guidance?



Personalized Recommendations

Tailored to each user's unique profile and goals.



24/7 Availability

Always on — no scheduling needed, no waiting.



Faster Decision-Making

Instant analysis and actionable career insights.



Interactive Conversations

Natural language dialogue for seamless user experience.

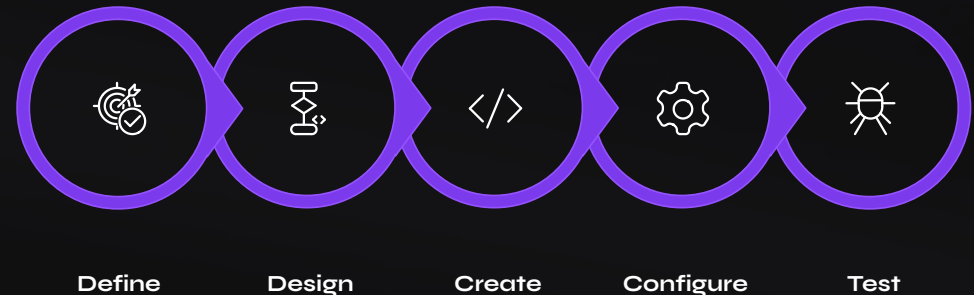


Scalable & Consistent

Serves thousands of students with uniform quality support.

AI Agent Development & Deployment

Step	Activity
1	Define the problem statement and project objectives
2	Design the AI agent conversation flow
3	Create HireMate AI in IBM watsonx Orchestrate
4	Configure description, behavior, and quick prompts
5	Test using sample user profiles
6	Refine responses and improve recommendations
7	Publish and deploy the AI agent
8	Generate and share the deployment link



Define

Design

Create

Configure

Test

Technologies Used & Project Outcomes

Technology Stack

Component	Technology
AI Platform	IBM watsonx Orchestrate
AI Model	IBM Granite Foundation Model
NLP	Natural Language Processing
Development	IBM SkillsBuild
Deployment	IBM watsonx Orchestrate

Project Outcomes



Personalized Career Guidance

Tailored career recommendations based on individual user profiles and goals.



Accurate Job Recommendations

Relevant job role suggestions aligned to user skills and industry demand.



Skill Gap Analysis

Clear identification of missing skills with targeted learning suggestions.



Enhanced Interview Preparation

Structured guidance to help students ace technical and HR interviews.



Improved Employability

End-to-end support that measurably boosts students' job readiness.

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HireMate AI

ProfileKnowledgeToolsetBehaviorChannels Preview

Profile

Describe what your agent does so people and other agents know when to use it.

Description*

HireMate AI is an intelligent career guidance agent that helps students and job seekers discover suitable job opportunities based on their skills, education, and interests. It provides personalized job recommendations, identifies skill gaps, suggests relevant certifications, offers resume improvement tips, and creates learning roadmaps to enhance career readiness and employability.

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

What makes a good description?

Model : GPT-OSS 120B — Ope...

View all

Build with AITalk to agent

Running in debug mode

Hello Ansley!

I'm glad you reached out. Based on the details you shared, I'll walk you through:

1. Job-role recommendations that align with your current profile and location preferences.

2. Skill-gap analysis for both a Software Developer track and an AI/ML Engineer track.

3. Targeted certifications & learning roadmap to close those gaps.

4. Resume-tuning tips (including a quick "before-and-after" example).

Save as test

Type something...

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Behavior

Define how and where your agent should react to requests and respond to users.

Instructions

HireMate AI is a professional and supportive career guidance assistant designed to help students, graduates, and job seekers make informed career decisions. It communicates in a friendly and respectful manner while collecting relevant information about the user's education, skills, interests, and career goals. The agent provides personalized job

Specify what your agent should do, how it should respond and any restrictions it should follow.

Guidelines

Add rules that guide an agent's actions and responses.

Add Guideline +

Build with AITalk to agent

Running in debug mode

intelligent career companion using...

Accuracy of generated answers may vary. Please double-check responses.

Recommend certifications and courses to boost my... →

Help me prepare for interviews with role-specific... →

< 1/3 >

Type something... →

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Build with AITalk to agent

Running in debug mode

10:29 PM

Hello, Welcome to HireMate AI-Your intelligent career companion using...

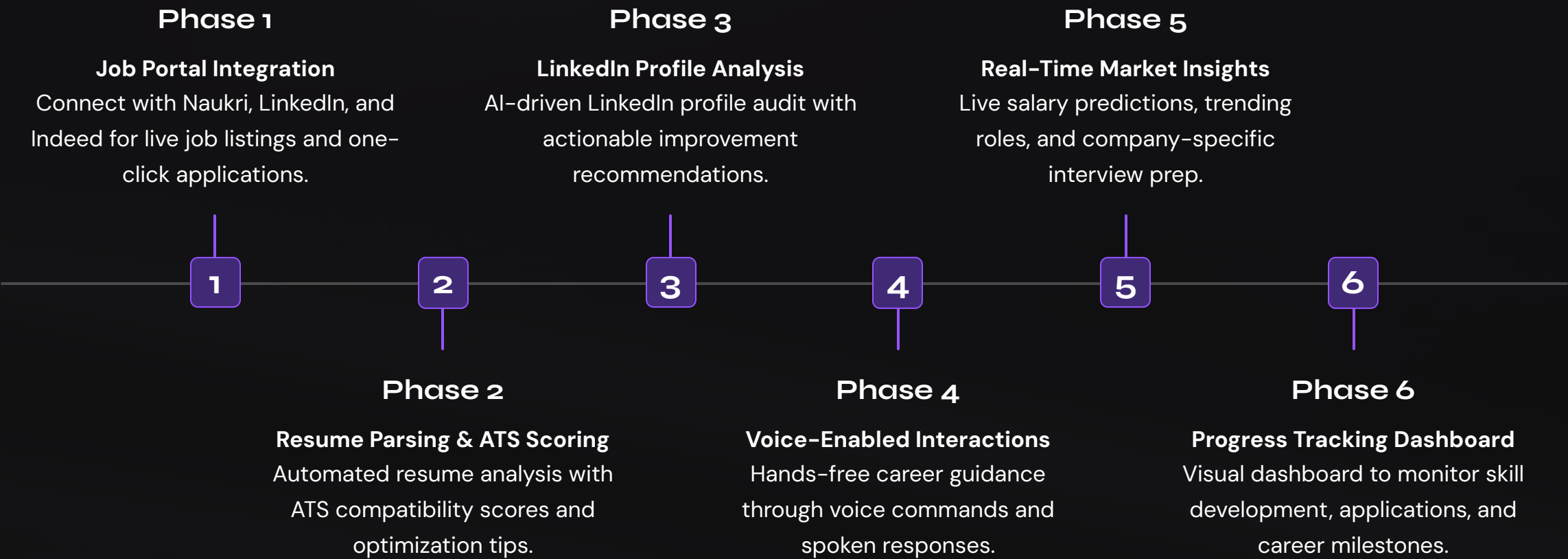
Accuracy of generated answers may vary. Please double-check responses.

Type something...

Monday 29 June 2026

Future Enhancements

HireMate AI is designed to evolve. The following enhancements will expand its capabilities, deepen personalization, and increase real-world impact for students and job seekers.



Conclusion

HireMate AI leverages the power of **Artificial Intelligence** to simplify career planning through personalized job recommendations, skill gap analysis, structured learning roadmaps, and comprehensive interview guidance — all delivered through a conversational AI agent built on **IBM watsonx Orchestrate**.

Personalized

Tailored career paths for every unique user

Impactful

Measurably improves employability outcomes

AI-Powered

Built on IBM Granite & watsonx Orchestrate

Scalable

Built to serve thousands of students seamlessly

Thank You